JK Lakshmipat University Jaipur
INSTITUTE OF ENGINEERING AND TECHNOLOGY

Mid Term Examination, October 2026

B.Tech., Semester-I, 2026-27

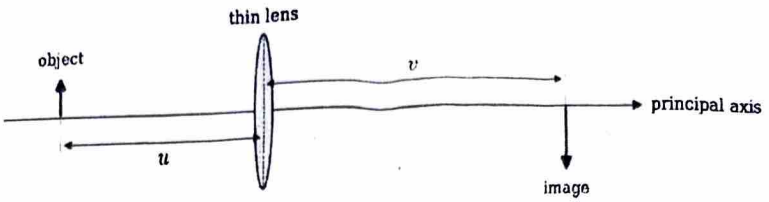
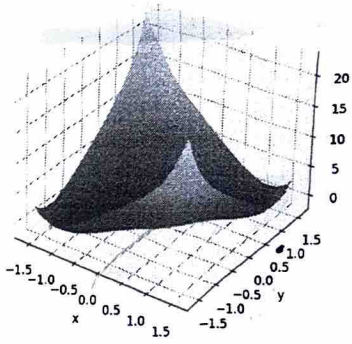
Roll No._____

AS1109: Calculus and Ordinary Differential Equations | Time: 2 hours | Max. Marks: 30

Instructions to students:

1. Do not write anything, other than your roll number, on the question paper.
2. Use of calculator is not permitted.
3. Marks will be awarded only for answers accompanied with correct justification.
4. If you encounter any ambiguity in a question, do not seek clarification from anyone. Make your assumptions, mention them clearly, and proceed with your solution.

Q 1.	(a) Suppose $f(x,y) \to L$ as $(x,y) \to (0,0)$ along every straight line through the origin. Does it follow that $\lim_{(x,y)\to(0,0)} f(x,y) = L$? Justify. (b) If f_x and f_y exist at a point, is $f(x,y)$ necessarily continuous there? Give justification. (c) Find all local maxima, local minima, and saddle points, if any, of $f(x,y) = e^{2x} + y^2$.	1+1+1 (CO 1, 2) BTL 2, 3
Q 2.	For $f(x,y) = xy$, find the range of f and sketch the level curves $f(x,y) = c$ for $c = 0, 1$.	2 (CO 1) BTL 1, 3
Q 3.	Evaluate the limit, if it exists (a) $\lim_{(x,y)\to(0,0)} \frac{x(x^2-y^2)}{x^2+y^2}$. (b) $\lim_{(x,y)\to(2,2)} \frac{x+y-4}{\sqrt{x+y}-2}$, $x + y \neq 4$.	2+2 (CO 1) BTL 3, 4
Q 4.	For the given function $h(x,y) = \frac{\ln(1-x^2-y^2)}{x-y}$ (a) Find and sketch the domain. (b) Find the points in $\mathbb{R}^2$ where the function is continuous.	2 (CO 1) BTL 3
Q 5.	Let $z = x^2 + y^2$, where $x = s + t$, $y = s - t$. Using the chain rule, find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ in terms of independent variables s and t.	2 (CO 1) BTL 3
Q 6.	Let $w = f(u) + g(v)$, where $u = x + 2y$, $v = x - 2y$. Show that $w_{xx} - \frac{1}{4} w_{yy} = 0$.	3 (CO 1) BTL 2,

Q 7.	Find the linear approximation $L(x, y, z)$ to $$f(x, y, z) = e^x + \sin(y + z)$$ at the point $P\left(0, \frac{\pi}{2}, 0\right)$.	2 *(CO 2)* *BTL 1, 3*
Q 8.	In a thin-lens model, the focal length F is related to the object distance u and image distance v by $$F(u, v) = \frac{uv}{u + v}.$$  At a particular setting, $u = 20$ cm and $v = 30$ cm. Each distance may have a measurement error of at most 0.1 cm. (a) Find $\partial F/\partial u$ and $\partial F/\partial v$, and using them write expression for the total differential dF. (b) Use differential to estimate the maximum absolute error in the computed focal length. (c) Which of the two measurements contributes more to the first-order uncertainty in F? Justify your answer.	2.5+1+0.5 *(CO 1, 2)* *BTL 2, 3, 4*
Q 9.	Consider $$f(x, y) = x^4 + y^4 - 4xy.$$ (a) Find all the critical points of f. (b) Compute $D = f_{xx}f_{yy} - (f_{xy})^2$ at every critical point. (c) Classify each critical point as a local maximum, local minimum, or saddle point. The surface $z = f(x, y)$ is shown in the accompanying figure.  	4 *(CO 1)* *BTL 2, 3*
Q 10.	(a) Find the length of the curve $y = \int_{-2}^{x} \sqrt{3t^4 - 1}\, dt$ from $x = -2$ to $x = -1$. (b) Find a curve through the point $(0, 1)$ whose length integral is $$L = \int_{-\pi/2}^{\pi/2} \sqrt{1 + \sin^2 x}\, dx.$$ How many such curves are there? Give reason for your answer.	2+2 *(CO 2)* *BTL 3*

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